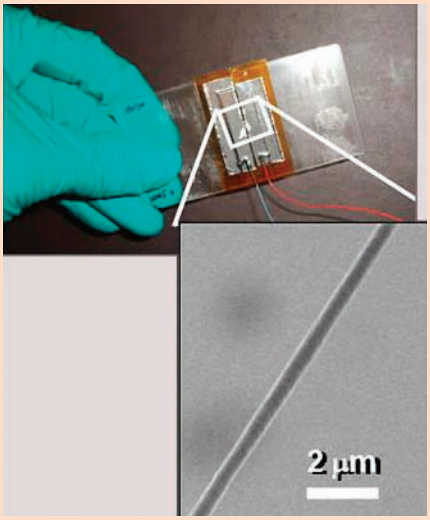
Adobe Flash 10.1 for mobiles
Adobe has launched Flash 10.1 for mobile phones and is available for immediate download for FroYo users for now

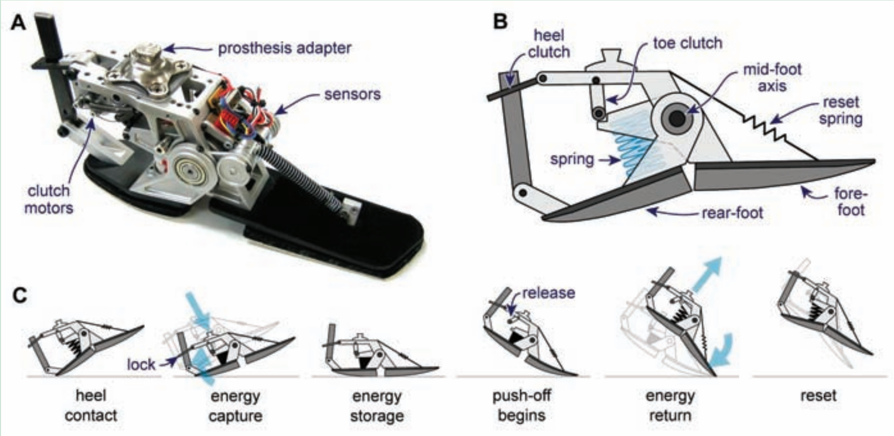
Google music download
Google plans to launch a music download service with online subscription service by 2011



A fibre nanogenerator on a plastic substrate

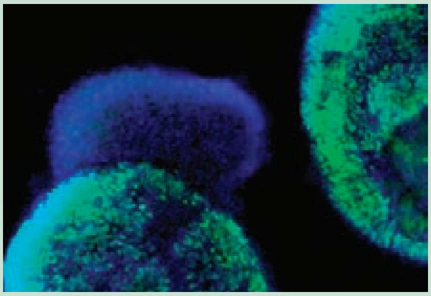
Nanotechnology
Clothes that charge
Ever had your MP3 player to run out of power just when your favourite song is about to come up? No sweat. Just plug it into your clothes. If you think we're kidding meet Mr. Liwei Lin, UC Berkeley professor of mechanical engineering and head of an international research team that has developed this new technology. The team has created fibre nanogenerators that have piezoelectric properties – fibres that can convert the stress and strain of stretches and twists into energy. Their success is thanks to a process they call "near-field electrospinning" – a technology that has enabled them to arrange the tiny fibres on a collector plate in a very controlled manner.

These fibres can be woven right into clothes without the least difference to the comfort of the wearer. This can have fantastic implications like being able to charge your cell phone while jogging or power up your laptop while you're exercising. Lin and his team have successfully transformed the mechanical energy of moving a finger into electricity by using the nanofibres attached to a surgical glove. Don't forget that a nanometre is just about the width of 10 atoms. It would take millions of such fibres to charge your laptop – but then again, a million such fibres would barely be bigger than a grain of sand. The availability of such garments could be a blessing to backpackers, soldiers, travellers and the like, who have to carry cumbersome chargers for their different gadgets, wherever they go.



Putting people back on their feet

Bio-tech
Prosthesis
The loss of a limb can have devastating effects and the only hope for the victim to lead a reasonably ordinary life again is a prosthetic limb – an artificial extension or replacement of the missing body part. Recent breakthroughs in prosthetic science have greatly improved the quality and range of prosthetic extensions. The use of electronics has augmented the positives too.



The green spheres are the early retinal cells

Artificial feet can be heavy and every amputee knows that it's a tough task to learn how to walk with prosthetic feet that have metal parts inside. An artificial foot created by Art Kuo (Art Kuo, professor of biomedical engineering and mechanical engineering at the University of Michigan) and Steve Collins helps the user to walk more easily by recycling the energy of each step. Although it's only in a prototype stage, the idea of an energy-recycling prosthetic foot is unprecedented. As the foot pushes off the ground, the machine inside the prosthetic limb catches the dissipated energy and returns it to the system at the appropriate moment with the help of a programmed microchip. This reduces almost by half, the amount of energy required by the user to lift the foot. The product is now being tested at the Seattle Veterans Affairs Medical Centre and is in the process of being commercialised.

Scientists at the University of California have created a 3D retina-like structure out of embryonic stem cells that have raised hopes of restoration of vision by the replacement of the retina by such bio-synthetic implants. A pilot test on a few patients has revealed that vision returns to a degree. Hans Keirstead, project head, along with his team, created two different types of cells from the stem cells taken from a human embryo – early cells of the retina and Retinal Pigment Epithelium cells. These were put together in an environment and exposed to different chemicals that encouraged them to grow. Remarkably, they grew to form three-dimensional structures.

Summing up
Arthur C. Clarke, in his many magnificent science fiction adventures, predicted the end of the world due to the consequences of technology several times. Hollywood movies such as the Terminator series, The Matrix and I, Robot, among many others, have often resorted to dystopian visions of technological development. At the end of the day, however, we only need to look around us to see the innumerable benefits of scientific development. Technology today is helping the blind to see and the crippled to walk. Power may become cheaper, robots assist us in our daily chores and printers print out things we need, when we need them. Technology, after all, is only as good as the hands it is in. As of now, it seems to be in safe hands.